



J. Ramulu Naik

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B.TECH

Electrical Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Indore	5.81 (Current)	2026-present
Senior Secondary	ssc Board	93.5 %	2022
Secondary	ssc Board	100.0%	2021

PROJECTS

- **HEALTH MONITORING AND RUL PREDICTION.** 2026 present
prof.Prathap reddy/-btp project Github
 - MATLAB based rul model is build to predict rul of 3 phase induction motor multi sensor are used to collect the signal data from motors
 - constructed a degradation based heath indicator and investigated rul estimation using data driven degradation modeling.
 - Designed the workflow for integration with IoT-based monitoring, enabling condition data acquisition, health assessment, and future maintenance alerts.
- **Buck Boost.** 2024 - Dec. 2025
prof.vijay/ project Github
 - Designed and implemented a buck boost converter with specification of input 20 volt ,output 15-25 volt swiching frequency 10 KHZ ,output current 1A. waveforms of switch voltage in CCM and diode current in CCM.
 - Reduce switching frequency to demonstrate DCM.
 - The project involved with selection ,circuit design, testing of pcb and simulation of waveforms to achieve a stable output. IT main focuses on the CCM and DCM mode by switching the frequency and duty circle.
- **GENETIC ALGORITHM.** 2024 - 2025
prof.Sharad kumar/ project Github
 - GENETIC ALGORITHM FOR KNAPSACK PROBLEM A genetic algorithm (GA) is a computational technique that uses principles of evolution and genetics to find solution to problem .
 - GAs are based on the idea of survival of the fittest. which states that the organisms which are most suited to their environment are more likely to survive and reproduce.
- **Line follower.** 2023 - apr. 2024
Omkar/Makerspace /project Github
 - Developed an automated vehicle that follows a visual black line or path on a surface.which controlled using infrared(IR) sensors and an arduino Uno.
 - The extra specification of the robort is to work like wireless charging with the help of sensors. and the robots wireless charging system uses advanced eletromagnetic induction technology.
 - It consistes of designing and fabricating of pcb, chassis,wheels. which involves from layout to testing.

TECHNICAL SKILLS

- **Programming:** PYTHON
- **softwares::** LTspice, PSbice, MATLAB simulink, EasyEDA, AutoCAD * Elementary proficiency

KEY COURSES TAKEN

- **CORE courses :** Basic Electrical Engineering,Basic Electronics Engineering,Digital system,Electrical machines, Electronic devices, ,Analog Circuit,Signals and systems,Basics power electronics,synchronous and special Electrical machine,restructure power electronics.
- **Mathematics:** Linear Algebra, Basic Calculus, Discrete Maths,probability &Random processes complex Analysis,differential equations numerical method.

POSITIONS OF RESPONSIBILITY

- **member,** AVANA Club, EBSB club. 2023 - 2025